Note 8 Autoregressive Models

1 Autoregressive (AR) Models

If a time series $\{X_t\}$ is generated from the following model

$$X_t = C + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + W_t,$$

where $\{W_t\}$ is white noise with mean zero and variance σ_W^2, and C is a constant, then $\{X_t\}$ is called an **autoregressive process of order p**, denoted as AR(p).

Remarks:

1. In AR models, every observation is a linear combination of the past observations and a error term.
2. Not all AR models are stationary.
3. For a stationary AR model, the constant C is not arbitrary.
4. It is usually convenient to assume normality of the white noise, that is,
$$X_t = C + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + Z_t,$$
where $\{Z_t\}$ is Gaussian white noise. This assumption is quite helpful in parameter estimation and inference.
5. Using the backshift operator, an AR(p) model can be written as

Specifically, an AR(1) process is

$$X_t = C + \phi_1 X_{t-1} + W_t$$

An AR(2) process is

$$X_t = C + \phi_1 X_{t-1} + \phi_2 X_{t-2} + W_t$$

2 Stationarity of AR(1) Process

3 Causality

If a time series $\{X_t\}$ can be expressed in the following form

$$X_t = C + \sum_{i=0}^{\infty} \psi_i W_{t-i},$$

with $\sum_{i=0}^{\infty} |\psi_i| < \infty$, then $\{X_t\}$ is said to be causal.

Remarks:

- All MA models are causal.

- An AR process is only stationary when it is causal.

Why is causality important?

3.1 Causality of AR(p)

Generally, a AR(p) process is causal if and only if all the roots of

$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p = 0$$

are outside the unit circle.

Example 3.1. Consider the AR(2) model below. Is it stationary? Is it causal?

$$X_t = 12 + \frac{5}{6} X_{t-1} - \frac{1}{6} X_{t-2} + W_t$$

3.2 ψ weights

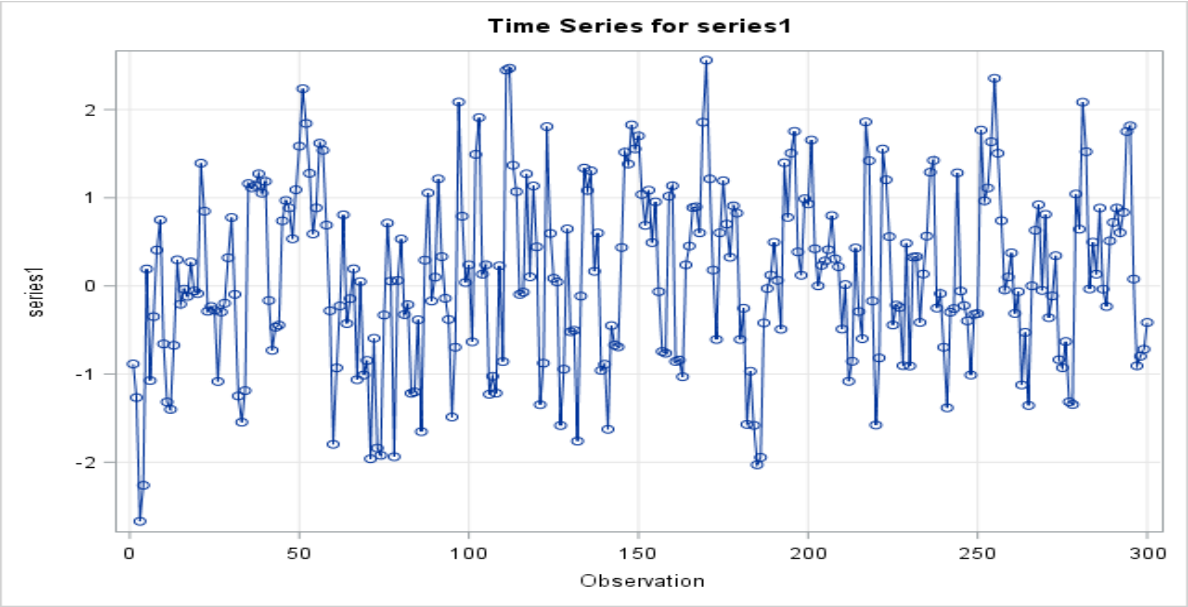
Example 3.2. We have shown in example 3.1 that the following AR(2) process is causal.

$$X_t = 12 + \frac{5}{6}X_{t-1} - \frac{1}{6}X_{t-2} + W_t$$

Find its ψ weights.

4 Model Identification

We are given a series $\{X_t\}$ of 300 observations. A plot of the series is shown below. We can see the series seems stationary. We want to use an AR model to fit the series. Which AR model should we use? Specifically, what is p?



4.1 Autocorrelation Function

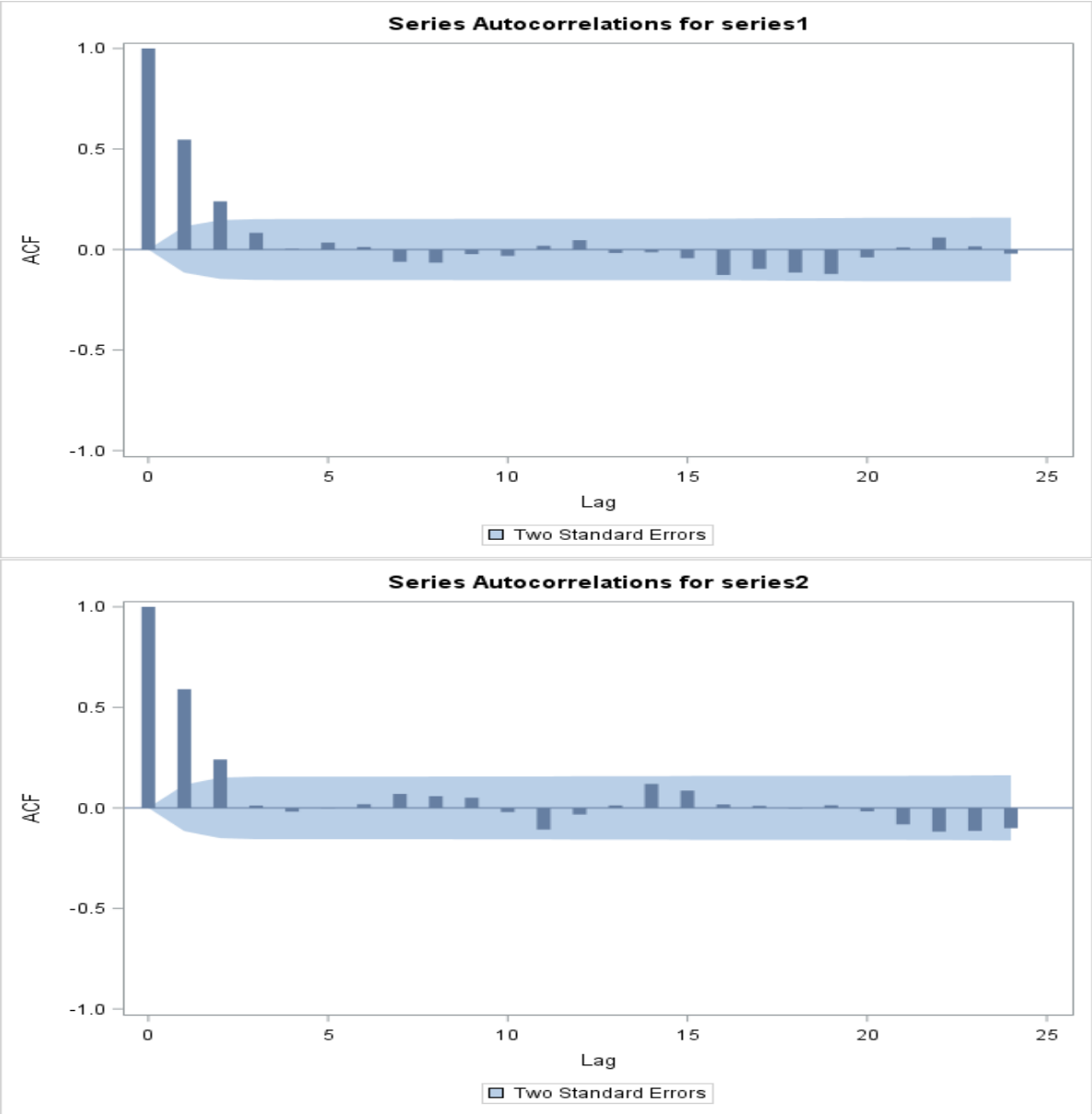
- ACF of AR(1)

Yule-Walker Equations:

- ACF of AR(p)

4.2 Partial Autocorrelation Function

Sometimes, it is not enough to only identify an AR model according to the ACFs. Consider another series $\{Y_t\}$, which is a MA(3). Both ACF plots of $\{X_t\}$ and $\{Y_t\}$ are shown below.



For a time series $\{X_t\}$, the **partial correlation** between X_t and X_{t+h} after adjusted for $X_{t+1}, \cdots, X_{t+h-1}$ is the conditional correlation

$$\begin{aligned}\alpha(h) &= \text{Corr}(X_t, X_{t+h}|X_{t+1}, \cdots, X_{t+h-1}) \\ &= \frac{\text{Cov}(X_t, X_{t+h}|X_{t+1}, \cdots, X_{t+h-1})}{\sqrt{\text{Var}(X_t|X_{t+1}, \cdots, X_{t+h-1})\text{Var}(X_{t+h}|X_{t+1}, \cdots, X_{t+h-1})}}\end{aligned}$$

Properties of partial autocorrelation function (PACF):

1. The PACFs $\alpha(h)$ are symmetric about zero, $\alpha(h) = \alpha(-h)$.

2. $\alpha(0) = \text{Corr}(X_t, X_t) = \rho(0) = 1$. Typically, we don't discuss $\alpha(0)$.

3. $\alpha(1) = \text{Corr}(X_t, X_{t+1}) = \rho(1)$.
4. In general, it is not easy to compute PACFs.
5. Fact: PACF of a stationary AR(p) process **cuts off** after lag p.

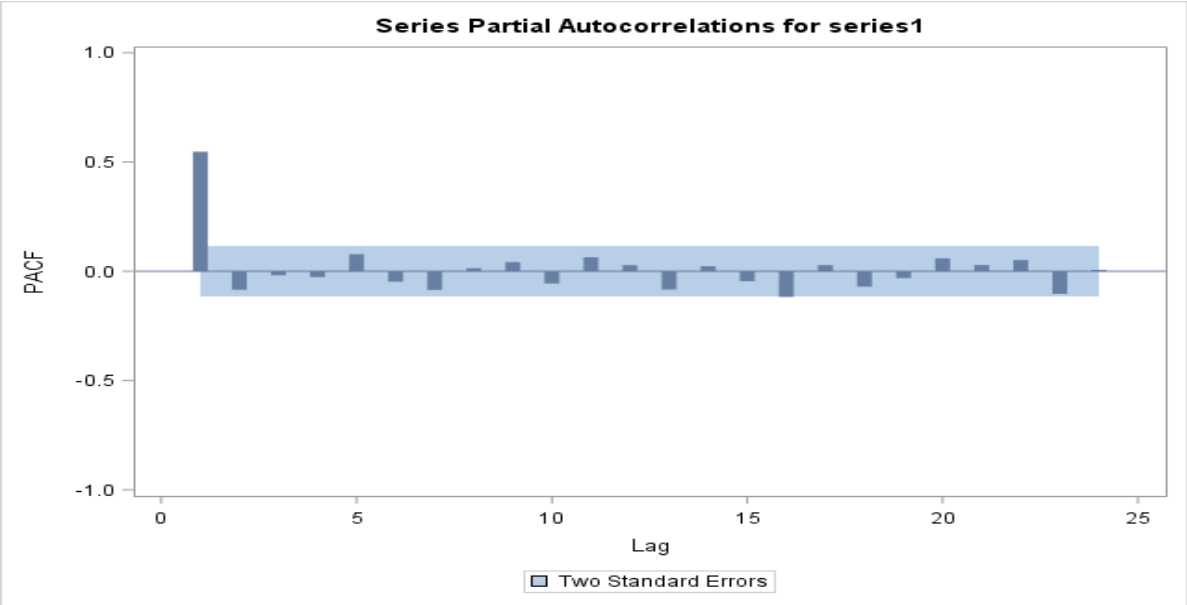
In application, PACFs $\alpha(h)$ are not observed from a given series. Instead, the $\alpha(h)$ are estimated by the sample PACF, $\hat{\alpha}(h)$.
Sample PACF are generally not easy to compute. In practice, we can use computing algorithms, such as Durbin-Levinson algorithm or innovation algorithm to simply the computation.

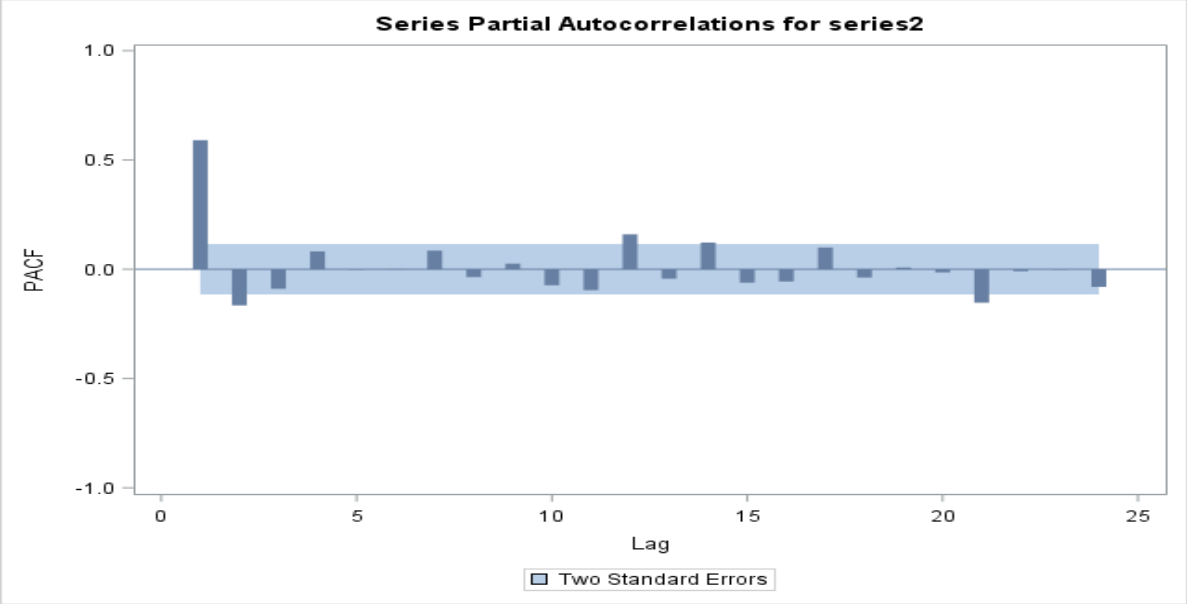
An asymptotic result of the sample PACF is

Let $\{X_t\}$ be a stationary time series with n observations, for n large enough,

$$\hat{\alpha}(h) \sim N\left(\alpha(h), \frac{1}{n}\right) \text{ approximately.}$$

With this result, we can have hypothesis tests.


Series Partial Autocorrelations for series1
PACF
1.0
0.5
0.0
-0.5
-1.0
0
5
10
15
20
25
Lag
Two Standard Errors


Series Partial Autocorrelations for series2
PACF
1.0
0.5
0.0
-0.5
-1.0
0
5
10
15
20
25
Lag
Two Standard Errors

5 Estimation

Once we have identified a series to be AR(p), we need to estimate its coefficients.

5.1 Maximum Likelihood Estimates

Let $\{X_t\}_{t=1}^n$ be a causal AR(p) process of length n,

$$X_t = C + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \cdots + \phi_p X_{t-p} + W_t,$$

if we assume $\{W_t\}$ is Gaussian white noise, then $\{X_t\}$ becomes a n-dimensional multivariate normal random variable with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Gamma}$, where

$$\boldsymbol{\Gamma} = \begin{bmatrix} \mathrm{Cov}(X_1, X_1) & \mathrm{Cov}(X_1, X_2) & \cdots & \mathrm{Cov}(X_1, X_n) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Cov}(X_2, X_2) & \cdots & \mathrm{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_n, X_1) & \mathrm{Cov}(X_n, X_2) & \cdots & \mathrm{Cov}(X_n, X_n) \end{bmatrix} = \begin{bmatrix} \gamma(0) & \gamma(1) & \cdots & \gamma(n-1) \\ \gamma(1) & \gamma(0) & \cdots & \gamma(n-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma(n-1) & \gamma(n-2) & \cdots & \gamma(0) \end{bmatrix}$$

The likelihood function of $\{X_t\}$ becomes

$$L(C, \phi_1, \cdots, \phi_p, \sigma_W^2 | X_1, \cdots, X_n) = \frac{1}{\sqrt{2\pi|\boldsymbol{\Gamma}|}} \exp\left(\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})' \boldsymbol{\Gamma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})\right)$$

Denote $\boldsymbol{\eta} = (C, \phi_1, \phi_2, \cdots, \phi_p, \sigma_W^2)$. The likelihood function can be rewritten as

$$L(\boldsymbol{\eta} | X_1, \cdots, X_n)$$

We obtain the estimate $\hat{\boldsymbol{\eta}} = (\hat{C}, \hat{\phi}_1, \hat{\phi}_2, \cdots, \hat{\phi}_p, \hat{\sigma}_W^2)$ by maximizing the L function.

$$\hat{\boldsymbol{\eta}} = \mathrm{argmax}\, L(\boldsymbol{\eta} | X_1, \cdots, X_n)$$

Example 5.1. Find the maximum likelihood estimates of $\{X_t\}$.

5.2 Unconditional Least Square Method

The unconditional least square (ULS) method is an alternation of the MLE that is computationally easier in the optimization step. Let $\boldsymbol{X} = (X_1, \cdots, X_n)'$ and

$$S(\boldsymbol{\eta}|\boldsymbol{X}) = \boldsymbol{X'\Gamma^{-1}X},$$

then maximizing the likelihood function is approximately equivalent to minimizing $S(\boldsymbol{\eta}|\boldsymbol{X})$.
The unconditional least square estimates are

$$\hat{\boldsymbol{\eta}} = \operatorname{argmin} S(\boldsymbol{\eta}|\boldsymbol{X})$$

The unconditional least squares estimates are obtained by minimizing the sum of squared residuals.

5.3 Conditional Least Square Method

The CLS estimates are conditional on the assumption that the past unobserved errors are equal to 0. For an AR(p) process, the conditional least square method obtains the estimates by minimizing

$$S(\boldsymbol{\eta}|\boldsymbol{X}) = \sum_{t=1}^{n} W_t^2 = \sum_{t=1}^{n}(X_t - \sum_{i=1}^{p} \phi_i X_{t-i})^2,$$

where the unobserved past values of X_t are set to 0.

5.4 Model Selection

6 Model Diagnostics

If the order p of the AR(p) is correctly identified and the parameters in the model are well estimated, the errors $\{W_t\}$ from the fitted model should be approximately white noise.
In practice, $\{W_t\}$ are not observed. We instead estimate W_t by the residual

$$R_t = X_t - \hat{X}_t,$$

where $\hat{X}_t$ is the fitted value at time t. For an AR(p) process,

$$\hat{X}_t = \hat{C} + \hat{\phi}_1 X_{t-1} + \cdots + \hat{\phi}_p X_{t-p}$$

Two assumptions are usually checked for the residuals:

- independence;
- normality (for MLE).

6.1 Independence Check

6.1.1 Ljung-Box Test

6.1.2 Individual Check

6.2 Normality Check

7 Forecast

7.1 Point Forecast

To forecast a future value, say $\hat{X}_t$, use the fitted model

$$\hat{X}_t = \hat{C} + \hat{\phi}_1 X_{t-1} + \cdots + \hat{\phi}_p X_{t-p}$$

- One-step ahead forecast:

- m-step ahead forecast:

Example 7.1. Predict X_{301} to X_{303} of the series $\{X_t\}$.

7.2 Prediction Interval